Turtle data format, as recorded by program TURTLE*.COM
(e.g., TURTLEP.COM, TURTLE23.COM, etc)

Last updated: July 1, 2025

Sample DAS section (starts below column indicators)

```
0        1         2         3         4         5         6         7
12345678901234567890123456789012345678901234567890123456789012345678901234567
-----------------------------------------------------------------------------
 35* 111100 090200 N36:46.37 W121:47.58
 36T.111107 090200 N36:46.48 W121:47.86   16
 37V.111107 090200 N36:46.48 W121:47.86    e    g    e    e    g
 38P.111107 090200 N36:46.48 W121:47.86   sb   jb   kf   td
 39A.111107 090200 N36:46.48 W121:47.86  650  100
 40W.111107 090200 N36:46.48 W121:47.86   rh   60    1    1    5
 41*.111200 090200 N36:47.78 W121:48.90
 42V.111207 090200 N36:48.01 W121:49.00    e    g    g    e    g
 43C.111234 090200 N36:48.60 W121:49.75 last transect was 14;
 44t.111250 090200 N36:48.84 W121:50.16   sb  -40   dc    6  190    n
 45*.111300 090200 N36:49.00 W121:50.45
 46S.111310 090200 N36:49.11 W121:50.67   17   kf   73    1   el
 47V.111324 090200 N36:49.42 W121:51.20    e    g    g    e    g
 48S.111326 090200 N36:49.46 W121:51.25   18   sb  -46   30   lo   zc
   1                                                          80   20
 49W.111328 090200 N36:49.49 W121:51.31    n   60    1    2    5
 50C.111340 090200 N36:49.66 W121:51.59 6 feet turtle
```

Columns	Item	Format
1-3	Event number	###
4	Event Code	#
5	Effort dot or blank	#
6-11	Time	HHMMSS
12	Blank	#
13-18	Date	MMDDYY
19	Blank	#
20-28	Latitude	NDD:MM.MM
29	Blank	#
30-39	Longitude	WDDD:MM.MM
40-44		
45-49		
50-54	Data fields. Information is event-code specific, according to key below	
55-59	All fields must be RIGHT JUSTIFIED within the 5 provided spaces.	
60-64		
65-69		
70+		

Event code	Col>40	Description/Key
* = Auto-position	--	Automatically logged position (every minute)
# = Deletion marker	--	Notes location of deleted entries.
C =Comment	41-132	Notes, corrections, molas, fish balls, etc.
E = End effort	--	Temporary end effort to circle, go over land/clouds etc.
R = Resume effort	--	Resume from temporary end effort

(Cont'd)

Event code	**Col 40+**	**Description/Key**
O= Transect End	--	Use to signal end of transect lines
T =Transect Start	40-44	Transect # (up to 4 numeric characters)
V =Viewing Condition		E=*excellent, G=Good, P=Poor, O=Off*
	40-44	Left inside (<35 degrees)
	45-49	Left outside (>35 degrees)
	50-54	Belly
	55-59	Right Inside (<35 degrees)
	60-64	Right Outside (>35 degrees)
P =Observer Codes		*2-character initials (unique)*
	40-44	Left Observer
	45-49	Belly Observer
	50-54	Right Observer
	55-59	Recorder
	60-64	Rear Left Observer (if present)
	65-69	Rear Right Observer (if present)
A = Altitude/Speed	40-44	Altitude in feet
	45-49	Speed in knots
W =Weather/Env.	40-44	*H=Haze/K=Kelp/R=Red tide/N=None*
		(Priority when more than one present: R > K > H > N)
	45-49	% overcast BETWEEN SUN AND VIEWING AREA
	50-54	Beaufort sea state *(0, 1, 2, 3, 4, 5)*
	55-59	Jellyfish *0=none, 1=few, 2=moderate, 3=lots*
	60-64	Horizontal sun *(Clock system, 12 = ahead, 6=behind*)
S = Sighting (Mammal)	40-44	Sighting number, numeric only, up to 4 digits
	45-49	Observer who made sighting
	50-54	Declination angle (LEFT = negative)
	55-59	Number of animals (best estimate)
	60-64	Species 1, 2-char. species code
	65-69	Species 2, 2-char. species code (blank if no other spp.)
	70-74	Species 3, 2-char. species code (blank if no other spp.)
1 = Ancillary sighting info		Species percentages, for multi-species sightings only
	60-64	Species 1 percent
	65-69	Species 2 percent
	70-74	Species 3 percent (blank if no other spp.)
s = Re-sight	40-44	Sighting number
	45-49	Declination Angle (LEFT=negative)
t = Turtle Sighting	40-44	Observer who made sighting
	45-49	Declination angle (LEFT = negative)
	50-54	Species code *(dc=leatherback, uh = unid. hard shell)*
	55-59	Size of turtle in feet

60-64	Travel direction in degrees *(0=North, 180=South)*
65-69	Tail Visible? *(Y=Yes, N=No, U=Unknown)*

SPECIES CODES:

Recorded using 'Sighting' key (F2)

Large whales

PM Sperm whale
MN Humpback whale
BM Blue whale
BP Fin whale
ER Gray whale
EG Right whale
BB Sei whale
BE Bryde's whale
UB Unid. baleen whale
LW Unid. large whale

Medium-sized whales

BD *Berardius bairdii*
ZI *Ziphius cavirostris*
UM *Mesoplodon* sp.
MC *Mesoplodon carlhubbsi*
UK *Kogia* sp.
BA Minke whale
SW Unid. small whale

Dolphins/Porpoises

PP Harbor porpoise
PD Dall's porpoise
DD *Delphinus* (unspecified)
DS *Delphinus* (short-beaked)
DL *Delphinus* (long-beaked)
LB *Lissodelphis*
LO '*Lags'* / Pacific white-sided
GG Grampus / Risso's
GM Pilot whale
OO Killer whale
UD Unid. dolphin/porpoise

Pinnipeds/Fissiped

PV Harbor seal
MA Elephant seal
EJ Steller sea lion
CU Northern fur seal
EL Sea otter
PU Unid. pinniped
US Unid. Seal
[ZC CA sea lion - Not recorded]

Other

M1 Small *Mola mola* (<2ft)
M2 Medium *Mola mola* (2-4ft)
M3 Large *Mola mola* (>4ft)

Recorded using 'Turtle Sighting' (shift-F4)

Turtles

DC Leatherback
CC Loggerhead
CM Green Turtle
UH Unid. hardshell

Miscellaneous (Recorded in comments)

ALBF Black-footed albatross
FB#x Fish Ball (e.g. fb1m, fb10s)
Sharks – comment
JFx### - JF species & % composition
(x = C for Chrysaora, M for moon jelly, E for egg-yolk jelly, and O for other)
(e.g. JFC080 JFM020 for 80% chrysaora, 20% moon jelly; record whenever species composition changes.)

DAS FILE NAMING CONVENTIONS FOR EACH SURVEY TYPE:

File root	
DASALL	
DASF#	Fine-scale leatherback surveys (#: M=Monterey Bay; A = Ano Nuevo; P = Pescadero Grid; F=Farallon Grid; FP = Farallon and Pescadero Grids; B = Bodega)
DASOW	Oregon/Washington
DASOFF or DASLUTH	Offshore surveys
DASNS	Non-standard surveys (no need to process)